

DU Mathsoc Problem Solving: Set 5

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~Advanced Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Let $a \geq 1$ be a real number. For any $n \in \mathbb{N}, n \geq 1$, $\lfloor a \rfloor^n \cdot \lfloor a^n \rfloor$ is a perfect square. Show that a is a natural number.

Problem 2. Prove that

$$\frac{1}{\sqrt{xyz(x+y+z)}} \geq \max \left\{ \frac{2}{(x+y)(y+z)}, \frac{2}{(x+z)(y+z)}, \frac{2}{(x+y)(x+z)} \right\},$$

for any positive real numbers x, y , and z .

Problem 3. Let p be a prime number, $p > 2$. Prove that p^2 divides $\left[\binom{np}{kp} - \binom{n}{k} \right]$, for any natural numbers n and k with $k \leq n$.

Problem 4. Let $a, b \in \mathbb{R}$ with $a < b$, and $f : [a, b] \rightarrow [a, b]$ a surjective function that is continuous on $[a, b]$ and differentiable on (a, b) . Prove that there exists $c \in (a, b)$ such that $(f'(c))^2 = 1$.

Problem 5. Show that there is no function $f : (0, +\infty) \rightarrow (0, +\infty)$ such that $f^2(x) \geq f(x+y) \cdot [f(x) + y], \forall x, y > 0$.